EMBEDDINGS & VECTOR SPACE CHEATSHEET

AI Engineer Ready · Week 2 — Turning Meaning into Numbers

Golden Rule → text → embedding vector → compare with cosine similarity → rank by closeness

1. What Is an Embedding?

An **embedding** is a list of numbers (a vector) that represents the **meaning** of a piece of text. Texts with similar meaning end up with vectors that are close together in that vector space.

```
$ "The cat sat on the mat." -> [-0.012, 0.087, 0.052, ..., 0.091] (e.g. 1536 numbers)
$ "The cat sat on the mat." -> [0.10, 0.20, 0.30, ...]
$ "A feline sat on a rug." -> [0.11, 0.19, 0.31, ...] <- close (similar meaning)
$ "Quantum computing rocks." -> [0.80,-0.10, 0.90, ...] <- far (different meaning)
```

2. Use Cases

Use case	What it means
Semantic search	Find documents similar in meaning to a query, not just matching keywords
Clustering	Group similar pieces of text together automatically
Recommendation	Suggest items whose embeddings are close to something the user liked
Anomaly detection	Flag text whose embedding sits far away from all the others

3. Choosing an Embedding Model

Scenario	Model	Why
Lightweight demo / prototype	all-MiniLM-L6-v2 (384-dim)	Fast and cheap, strong enough for early-stage demos
Production search / RAG	all-mpnet-base-v2 (768-dim)	Better semantic coverage and retrieval quality
High-stakes retrieval	OpenAI embedding models (API)	Highest accuracy for ambiguous or high-precision text

4. Cosine Similarity

```
$ cos_sim(a, b) = (a . b) / (||a|| * ||b||)
$ from sklearn.metrics.pairwise import cosine_similarity
$ score = cosine_similarity([vec_a], [vec_b])[0][0]
$ # 1.0 = identical direction (very similar meaning)
$ # 0.0 = unrelated | -1.0 = opposite meaning
```

5. Generating an Embedding (OpenAI)

```
$ from openai import OpenAI
```

```
$ client = OpenAI()
$ resp = client.embeddings.create(
$ model="text-embedding-3-small",
$ input="The cat sat on the mat.",
$ )
$ vector = resp.data[0].embedding
```

Pitfall — comparing embeddings from different models

Never compute similarity between a vector from one embedding model and a vector from another — different models produce different vector spaces (and often different dimensions), so the distances are meaningless across models.